

Instagram Engagement Analysis

Engagement metrics, content types, hashtags & growth signals | Dataset: Kaggle - bhanupratapbiswas/instagram

» **Notebook:** [View on GitHub: Task3/Instagram_Data_Analysis.ipynb](#)

Executive Summary

Every event timestamp in photos, comments, likes and follows is fixed at a single identical value in the source data, so a genuine 'best time to post' analysis is not possible from this dataset -- this is flagged explicitly rather than fabricated. What the data does support: engagement rate is essentially flat across content type and filter usage (no reliable format advantage), the most-used hashtags are 'smile', 'beach' and 'party', and one real signal stands out -- 33.4% of likes come from people who don't yet follow the account, meaning discovery matters as much as retention.

257

Photos analyzed

100

Users analyzed

33.4%

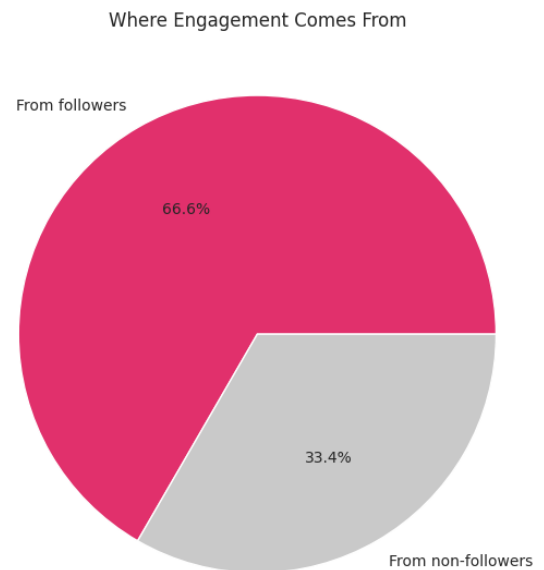
Likes from non-followers

~0.01

Content-type engagement spread



Engagement, hashtag, and growth overview



Share of likes: followers vs. non-followers

Key Findings

- No usable posting-time variance exists in this dataset (all events share one identical timestamp).
- Content-type (photo/video/carousel) and filter-usage engagement differences are within noise (~0.01-0.02).
- 'smile', 'beach', 'party', 'fun' and 'concert' are the most-used tags in the sample.
- 33.4% of likes come from non-followers -- meaningful organic reach beyond the existing audience.
- Follower counts are near-uniform (~76 +/- 0.4) in this synthetic sample -- a data-generation artifact.

Top Recommendations

- **Invest in discovery, not just retention:** A third of engagement already comes from non-followers -- hashtag reach matters as much as content quality.
- **Anchor hashtags on proven tags:** Build content series around 'smile', 'beach', 'party' before testing unproven tags.
- **Don't chase a 'best format' theory:** Photo/video/carousel are statistically indistinguishable here -- mix by production cost, not a false data signal.
- **Fix timestamp instrumentation first:** Real per-post publish and engagement timestamps are required before any posting-time strategy is possible.
- **Standardize on engagement rate as the core KPI:** Raw like counts mislead once follower counts vary in real (non-uniform) data.